

1. Administrative & Study Identification

Full Study Title: A Phase 4, Multicenter, Randomized, Double-blind, Parallel Group, Placebo Controlled Study to Evaluate the Effect of Benralizumab on Structural and Lung Function Changes in Severe Eosinophilic Asthmatics **Study Title:** Benralizumab Airway Remodeling Study in Severe Eosinophilic Asthmatics **Short Title/Acronym:** CHINOOK **Protocol Number:** D3250C00059 **NCT Number:** NCT03953300 **Posting ID:** 1051891555610447130 **Sponsor/Company Name:** AstraZeneca **Investigational Product:** Benralizumab (Fasenra, ATC Code: R03DX10) / Placebo **Phase of Development:** Phase 4 **Trial Status:** Active, Not Recruiting **Disease Areas:** Asthma (Eosinophilic Asthma) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effect of benralizumab on submucosal eosinophils in endobronchial biopsies and to evaluate the effect of treatment with benralizumab on airway wall dimensions as measured by quantitative computed tomography (qCT) imaging up to Week 48. **Secondary Objectives:** To evaluate the effect of benralizumab on small airway obstruction and large airway remodeling, and to assess the association between changes in airway wall structure and lung physiology measures (including gas trapping and airway resistance).

2.2 Background & Rationale To address the mechanistic gap in understanding how sustained depletion of airway eosinophils impacts airway remodeling. While Phase 3 trials have previously demonstrated benralizumab's ability to significantly reduce annual exacerbation rates and improve lung function, this mechanistic study evaluates whether macroscopic, morphological, and structural changes in the bronchial wall are effectively modulated by benralizumab in patients with Severe Eosinophilic Asthma (SEA) whose symptoms persist despite standard background therapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (Parallel Group, 2:1 ratio)

3.2 Planned Enrollment & Duration Target **NS:** 74 **Number of Sites:** Multicenter **Estimated Study Start:** 08-Oct-2019 **Estimated Primary Completion:** 19-Oct-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female aged 18 through 70 years. 2. Physician-diagnosed asthma requiring continuous treatment with medium- or high-dose inhaled corticosteroids (ICS) plus a long-acting B2-agonist (LABA) with or without additional controller medication for at least 12 months prior to Visit 1, and current treatment with high-dose ICS plus LABA for at least 3 months prior to Visit 1 with or without additional asthma maintenance medication. 3. Morning pre-BD FEV1 $\geq 50\%$ to $< 80\%$ of predicted normal value (PNV) and ≥ 1 liter (L) or morning pre-BD FEV1 $\geq 50\%$ to $< 90\%$ of PNV, if historical pre-BD FEV1 value (within 12 months prior to screening visit) was $< 80\%$ of PNV. 4. A blood eosinophil count meeting any of 3 criteria: ≥ 300 cells/ μ L during screening or ≥ 220 to < 300 cells/ μ L during screening and documented eosinophil count of ≥ 300 cells/ μ L in the past 12 months, or ≥ 150 to < 300 cells/ μ L during screening plus one of the following: presence of nasal polyps or pre-BD FVC $< 65\%$ predicted at Visit 2. 5. Negative pregnancy test. 6. Asthma control questionnaire (ACQ-6) > 1.5 . 7. Fewer than 12 exacerbations within the 6 months prior to Visit 3.

4.2 Exclusion Criteria 1. Any disease or concomitant medication which could affect study results or safety of study participants, including: current smokers, history of cancer, life-threatening asthma, and clinically important pulmonary disease other than asthma. 2. Use of chronic immunosuppressive medication or receipt of immunoglobulin (or blood products) within 30 days prior to the date informed consent is obtained. 3. Previously received benralizumab, live attenuated vaccines 30 days prior to the date of randomization, bronchial thermoplasty in the last 24 months prior to Visit 1, any investigational non-biologic within 30 days (or 5 half-lives) prior to consent, or any marketed or investigational biologic for the treatment of asthma within 4 months (or 5 half-lives) prior to consent. 4. Currently pregnant, breastfeeding, or lactating women.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Benralizumab 30 mg/mL (or matching placebo) administered every 4 weeks for the first 3 doses (Weeks 0, 4, 8), followed by administration every 8 weeks thereafter (Weeks 16, 24, 32, 40). **Route of Administration:** Subcutaneous (SC) injection via accessorized prefilled syringe (APFS). **Duration of Treatment:** 48 weeks of exposure, followed by a 4-week follow-up period.

5.2 Concomitant & Prohibited Therapy Permitted: Maintenance background therapy including ICS plus LABA, with or without additional asthma controller medications. **Prohibited:**

Investigational non-biologics within 30 days, live attenuated vaccines within 30 days prior to randomization, and any marketed/ investigational biologics for asthma treatment within 4 months prior to consent.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, Eosinophil Labs, ACQ-6
Baseline	Day 0	Randomization, Baseline qCT & biopsies, First SC Dose
Interim	Weeks 4, 8, 16, 24, 32, 40	Subcutaneous Dosing, Safety Labs, Lung Physiology Testing (Spirometry/Oscillometry)
End of Study	Week 48 & 52	Week 48 Final qCT and Biopsies; Week 52 Final Safety Follow-up, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from baseline in submucosal eosinophil count in endobronchial biopsies and change in airway wall dimensions up to Week 48. **Measurement Method:** Major Basic Protein (MBP) staining for biopsies and Quantitative Computed Tomography (qCT) for airway wall dimensions.

7.2 Secondary & Exploratory Endpoints * Change from baseline in small airway obstruction and large airway remodeling parameters. * Association between changes in airway wall structure and lung physiology measures (e.g., gas trapping via whole body plethysmography and airway resistance).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Method of collection includes site HCP interview, standard lung function testing, and clinical laboratory checks at each dosing visit. **Serious Adverse Events (SAEs):** Reporting timeline is 24 hours of site awareness to the sponsor safety team. **Data Monitoring Committee (DMC):** Internal sponsor medical review committee oversight for safety trend monitoring.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy evaluation; Safety Set for all adverse events. **Sample Size Justification:** Target enrollment of N=74\$, randomized 2:1, to achieve sufficient power ($\alpha = 0.05$) to evaluate histological and structural changes in target variables. **Handling of Missing Data:** Standard Mixed-Model Repeated Measures (MMRM)

and/or Multiple Imputation (MI) applied to longitudinal qCT and histological datasets.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite visits and periodic remote source data verification by clinical research associates (CRAs).

Audit Trail: Standardized electronic case report form (eCRF) locking procedures, query generation, and centralized review for all qCT imaging and biopsy samples.

11. Study Identifiers & Governance

Primary ID: D3250C00059 **Posting ID:** 1051891555610447130 **Registry Number:** NCT03953300 **Trial Status:** ACTIVE_NOT_RECRUITING **Sponsor/Lead Organization:** AstraZeneca **Study Title:** Benralizumab Airway Remodeling Study in Severe Eosinophilic Asthmatics (CHINOOK) **Official Regulatory Title:** A Phase 4, Multicenter, Randomized, Double-blind, Parallel Group, Placebo Controlled Study to Evaluate the Effect of Benralizumab on Structural and Lung Function Changes in Severe Eosinophilic Asthmatics

12. Clinical Framework (Asthma Specific)

Primary Condition: Severe Eosinophilic Asthma (SEA) **Disease Area:** Asthma (Eosinophilic Asthma) **Diagnosis Threshold:** Blood eosinophil count ≥ 300 cells/ μ L & ACQ-6 ≥ 1.5 **Medical Methodology:** Standard background therapy (ICS + LABA) **Optimization Target:** Airway tissue eosinophil depletion and reversal of airway wall remodeling

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for SEA management combined with IL-5R α targeted biologic therapy. **Education**

Protocol (HCP): Site-level training focused on qCT imaging standards and endobronchial biopsy procedures. **Education Protocol**

(Patient): General asthma self-management, exacerbation tracking, and inhaler technique reinforcement. **Quality Audit Mechanism:** Centralized evaluation and scoring of histological stains (MBP) and respiratory scans to limit variability.

14. Observation & Follow-Up Schedule

Total Duration: 52 Follow-up weeks (48 weeks treatment + 4 weeks off-treatment follow-up) **Onsite Visit Cadence:** Baseline, Weeks 4, 8, 16, 24, 32, 40, 48, and Week 52 **Remote Monitoring Frequency:** Intermittent phone contacts for exacerbation and AE screening between 8-week dosing intervals **Checklist Review Interval:** Routine eCRF review matching the dosing cadence

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					